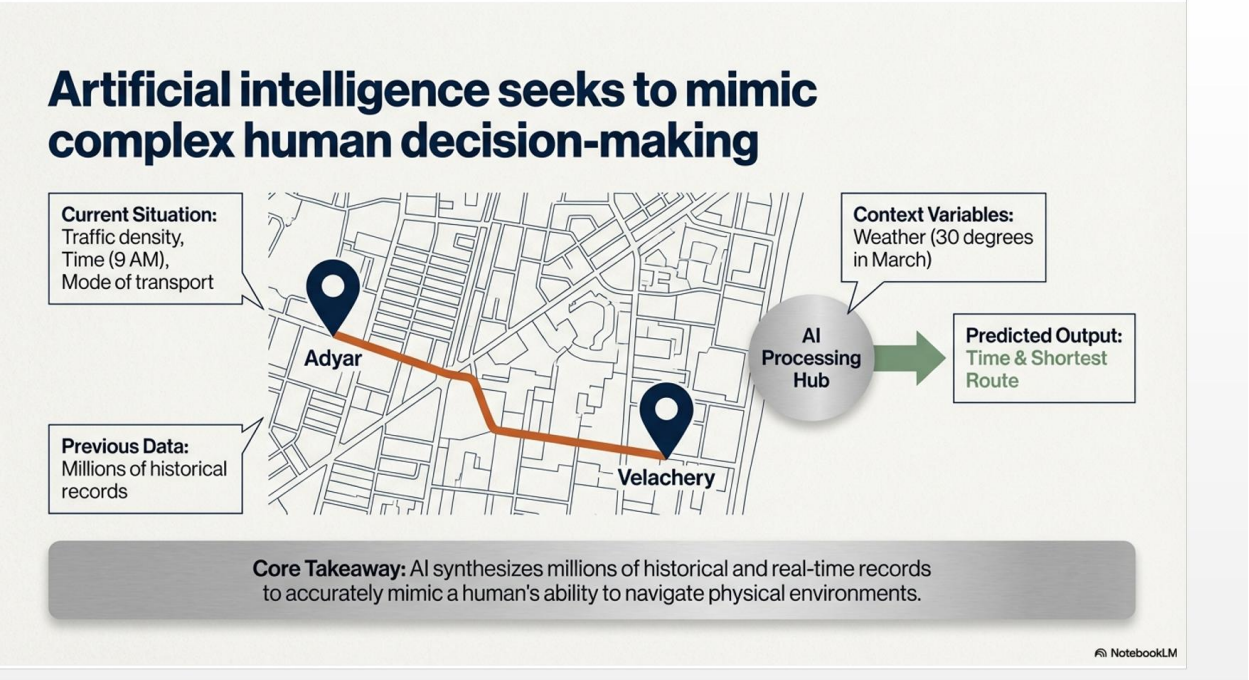
Creation of a PowerPoint using the Agenda file



The capabilities of AI form a nested ecosystem of increasing complexity

Artificial Intelligence (Mimic)

The overarching concept of intelligent systems.

Machine Learning (Predict)

Systems trained on thousands to millions of records.

Deep Learning (Scale)

Neural networks processing millions to trillions of records.

Generative AI (Create)

Multimodal content generation and net-new outputs.

NotebookLM

Machine Learning operates on the foundational principle of prediction of prediction

Core Function: Predictive AI

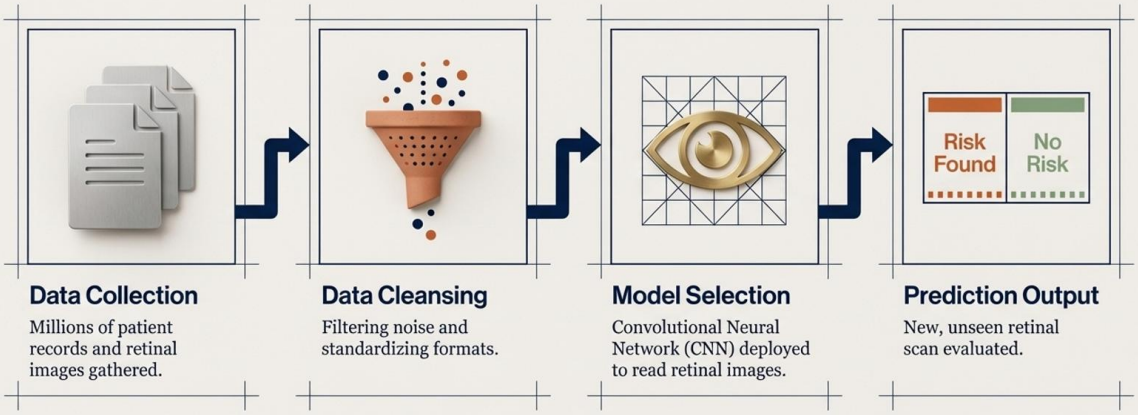
Data Scale: Requires thousands to millions of records.

Primary Use Case: Predicting Heart Disease Risk

Name	Age	Blood Group	Diabetic	Blood Pressure	Location	Genetic History	Smoker	Drinker	Retinal Image

NotebookLM

Turning raw data into predictive intelligence requires a structured pipeline



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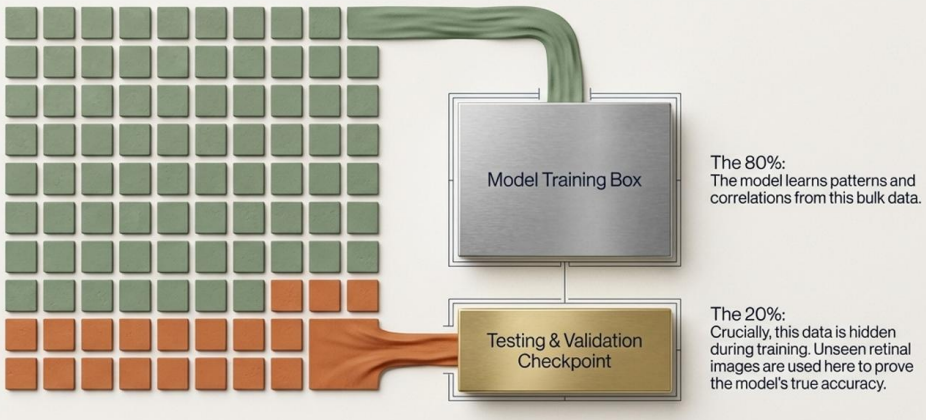
Models must be validated on data they have never seen before



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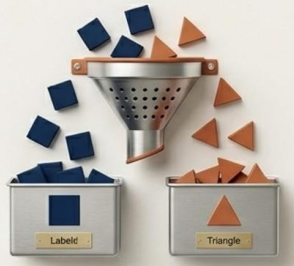
Algorithms learn differently depending on the presence of existing labels

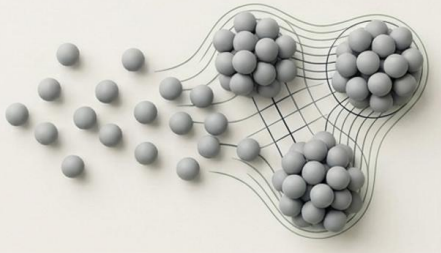
Supervised Learning



Data Type:
Labelled Data (e.g., Content-based star ratings, explicit likes/dislikes).

Core Function:
Predicting outcomes or categorizing.

Unsupervised Learning



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NotebookLM

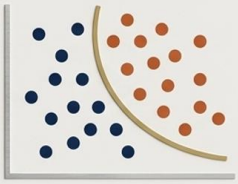
Supervised Learning

Linear Regression

Output Type: Numbers / Numerical Values

Real-World Examples:
- Predicting home purchase prices in real estate.
- Predicting the exact number of software defects in testing.

Classification



Output Type: Labels / Categories

Real-World Examples:
- Google predicting the risk of a heart attack (Yes or No).
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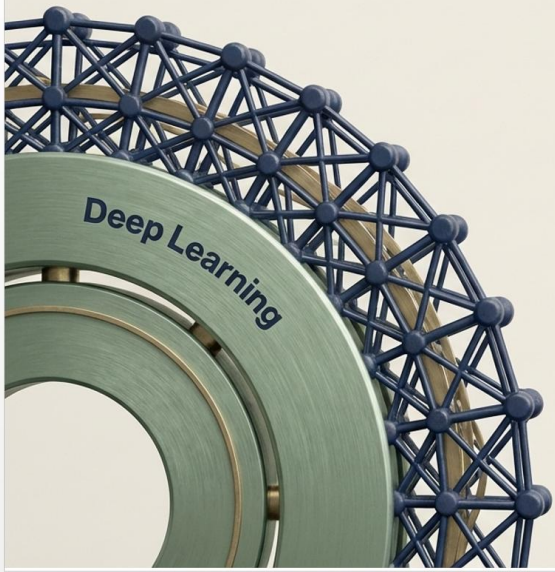
NotebookLM

Deep Learning applies neural networks to massive datasets



The Equation

Machine Learning + **Vast Amount of Data** = **Deep Learning**

The Scale Jump

Machine Learning: 1,000s to Millions of records

Deep Learning: Millions to Trillions of records

Functionality

Core Function: Remains fundamentally Predictive AI, but engineered for vastly more complex, non-linear pattern recognition.

NotebookLM

Generative AI represents a fundamental shift from prediction to creation

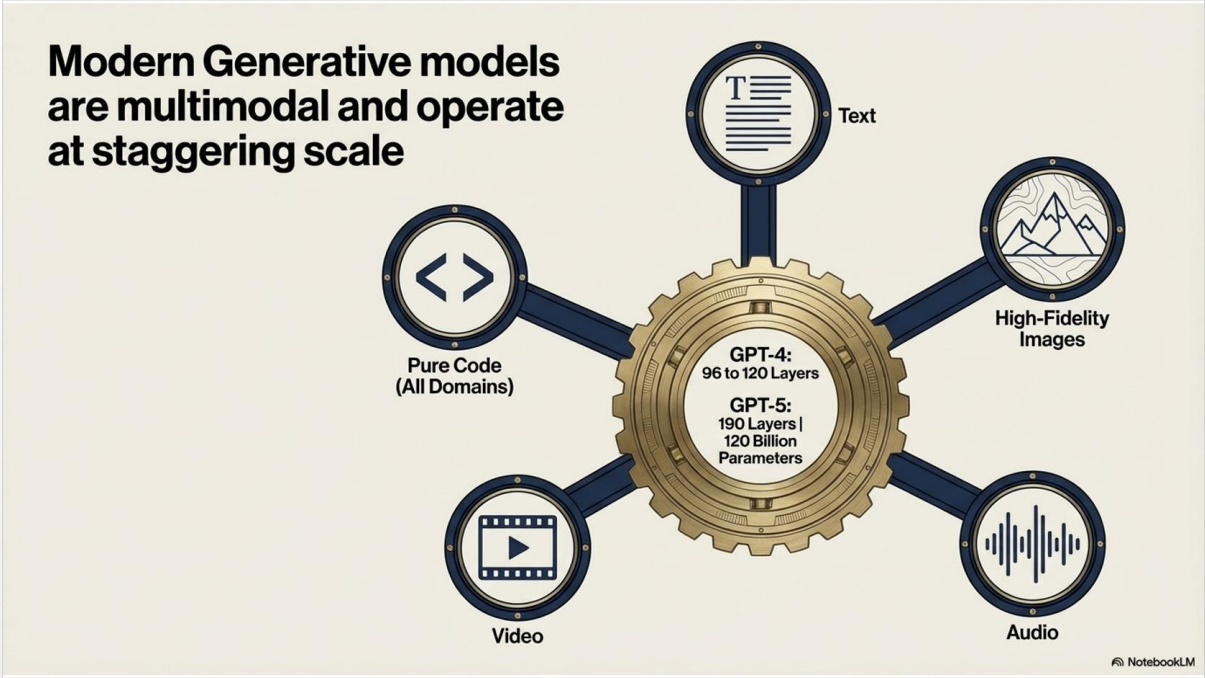
Taxonomy: A direct, specialized subset of Deep Learning.

The Pivot: Explicitly Not Predictive. Its sole function is to generate net-new, original content.

The Engine: Powered by LLMs (Large Language Models) developed by providers like OpenAI.

Definition: GPT stands for Generative Pretrained Transformer. These models are pretrained on vast, internet-scale datasets spanning all human fields, not just software.

NotebookLM



The Intelligence Ecosystem Synthesized

	Machine Learning	Deep Learning	Generative AI
Primary Function	Predictive	Predictive (Complex)	Generative / Creative
Architecture & Engine	Statistical Algorithms	Neural Networks	Pretrained Transformers (LLMs)
Data Scale	1,000s to Millions of records	Millions to Trillions of records	Billions of Parameters / Internet-scale
Target Outputs	Numbers, Categories (Yes/No)	Complex classifications	Text, Image, Audio, Code

NotebookLM

Pdf format generated using the same file in notebook LLM

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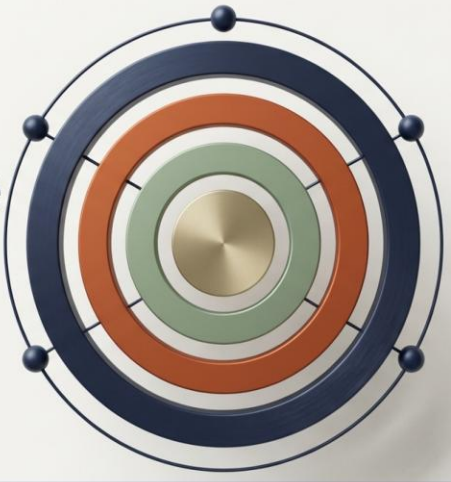
Demystifying the Intelligence Ecosystem

From foundational predictions to generative frontiers.



Day 1 Foundations

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Artificial intelligence seeks to mimic complex human decision-making

Core Takeaway: AI synthesizes millions of historical and real-time records to accurately mimic a human's ability to navigate physical environments.

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The capabilities of AI form a nested ecosystem of increasing complexity

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—	—	—	—	—	—	—	—	—	—
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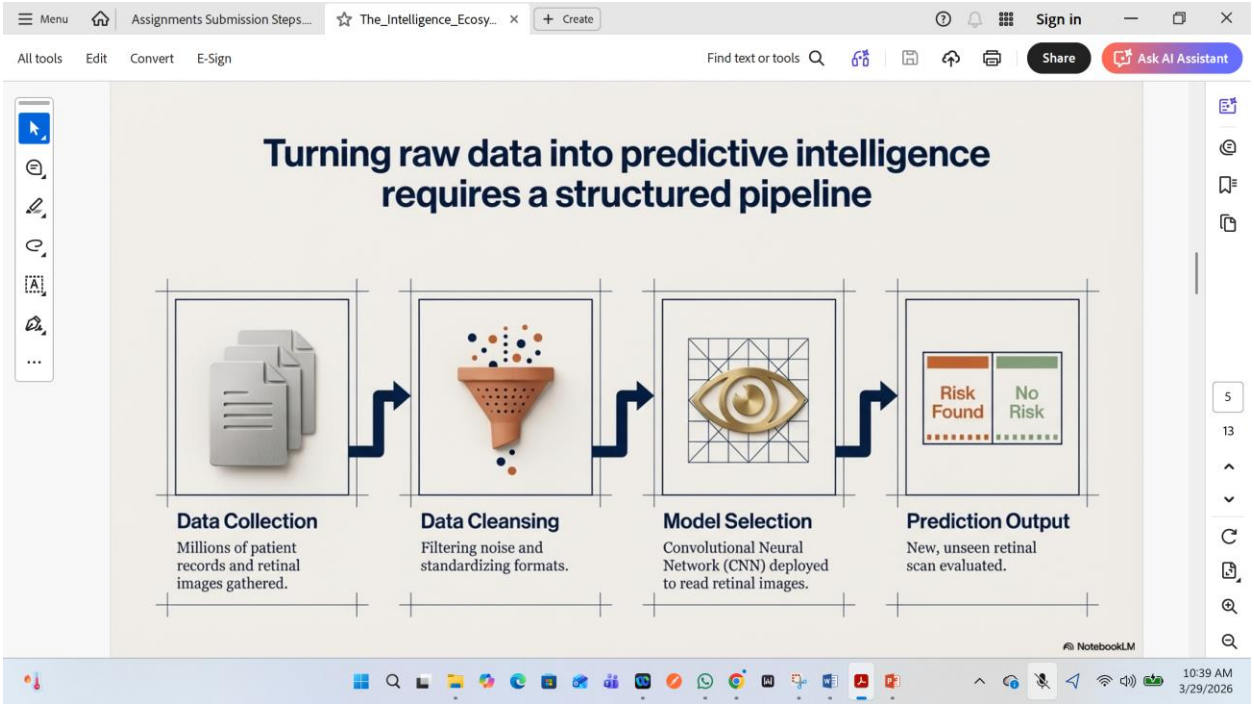
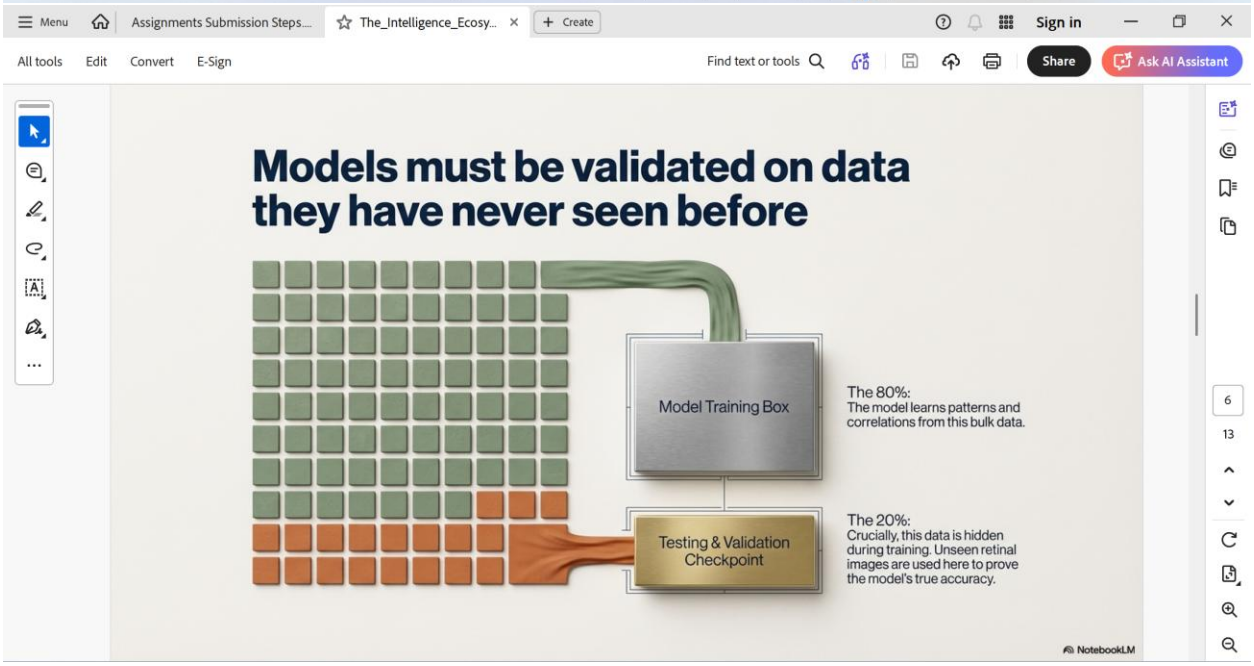
NotebookLM


Turning raw data into predictive intelligence requires a structured pipeline
Risk Found
No Risk
Data Collection
Millions of patient records and retinal images gathered.
Data Cleansing
Filtering noise and standardizing formats.
Model Selection
Convolutional Neural Network (CNN) deployed to read retinal images.
Prediction Output
New, unseen retinal scan evaluated.
NotebookLM


Models must be validated on data they have never seen before
Model Training Box
Testing & Validation Checkpoint
The 80%:
The model learns patterns and correlations from this bulk data.
The 20%:
Crucially, this data is hidden during training. Unseen retinal images are used here to prove the model's true accuracy.
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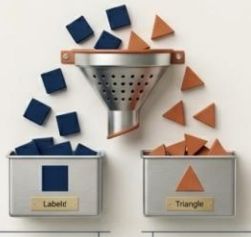
Algorithms learn differently depending on the presence of existing labels

Supervised Learning



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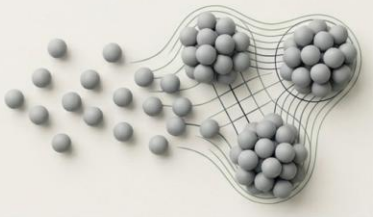

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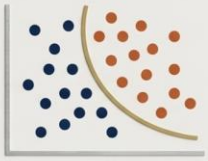

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Ask AI Assistant Why is this information important?
By using AI Assistant, you agree to Generative AI User Guidelines.
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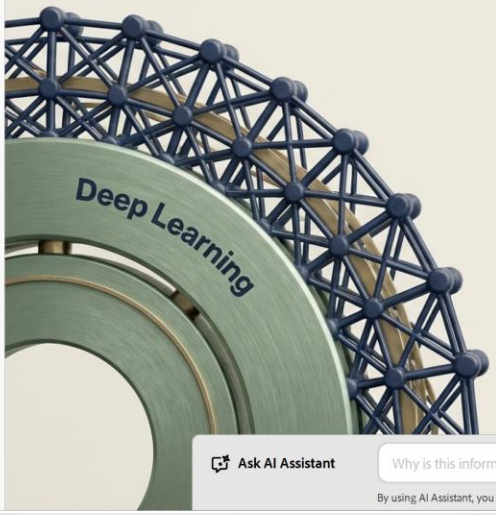

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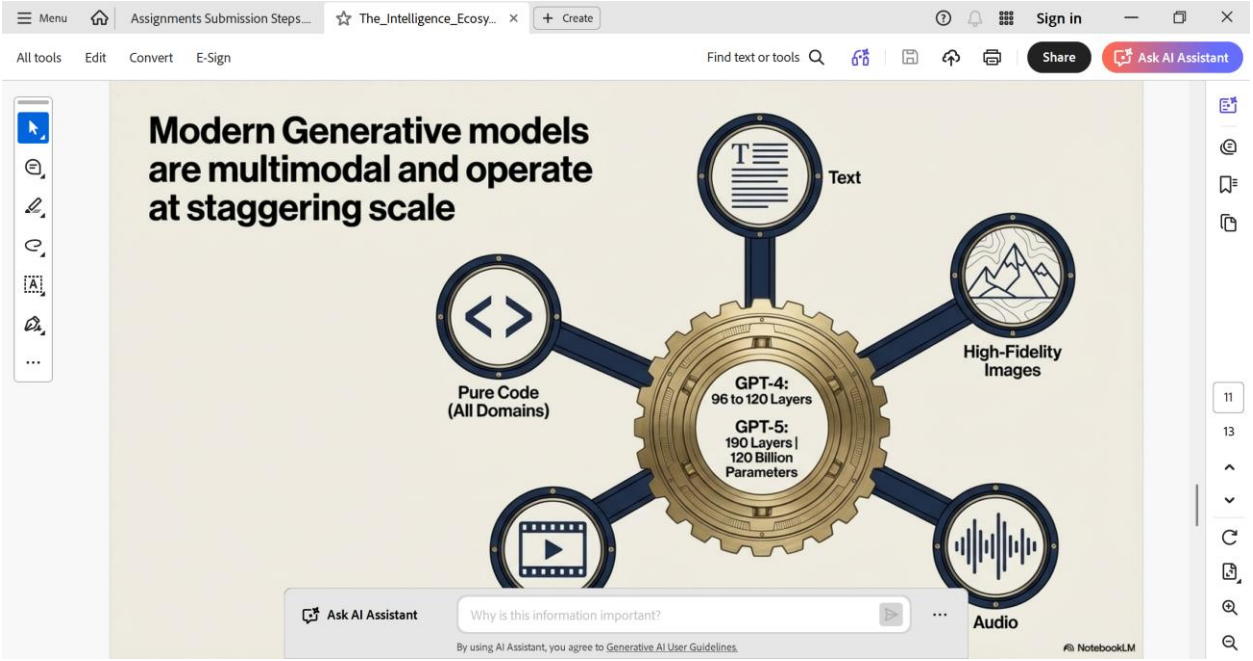

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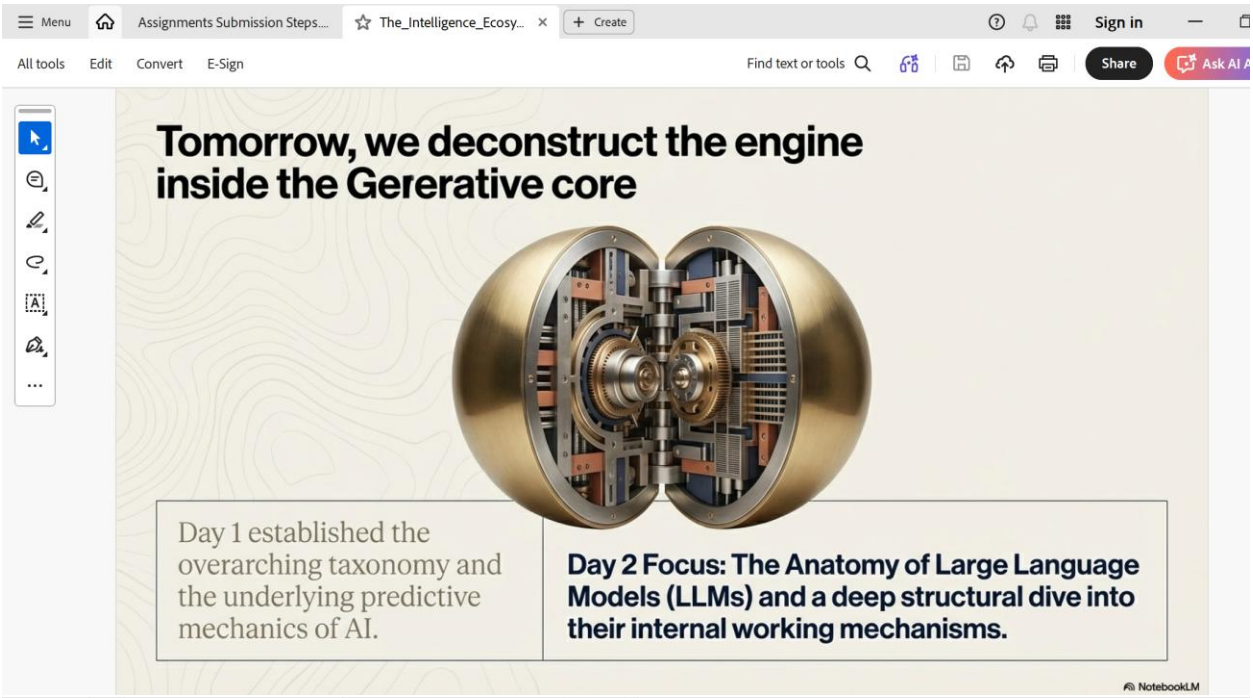

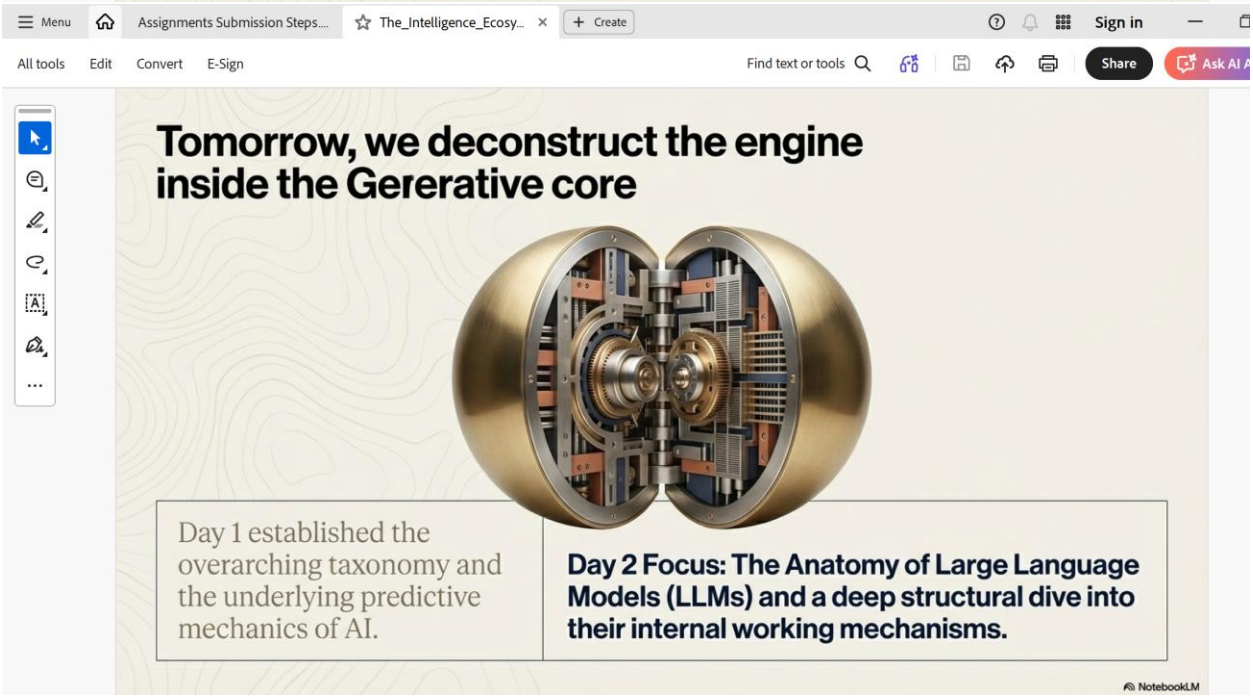

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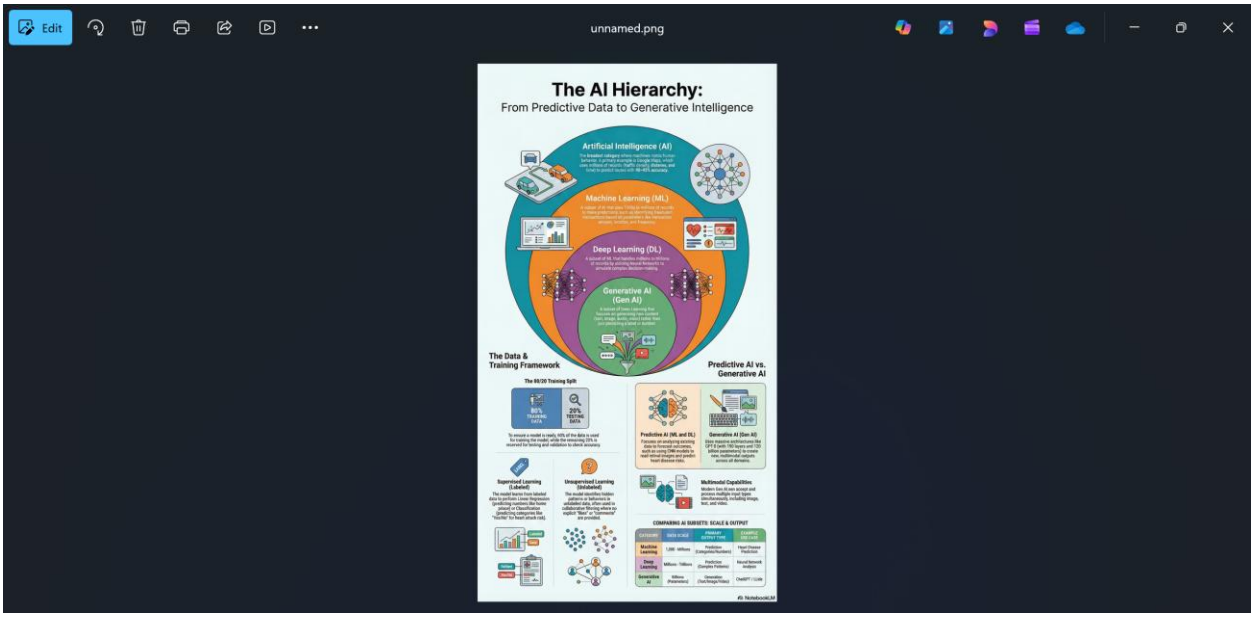
Edit
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The AI Hierarchy:
From Predictive Data to Generative Intelligence
Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)
Generative AI (Gen AI)
The Data & Training Framework
Predictive AI vs. Generative AI
The 80/20 Training Split
80% TRAINING DATA
20% TESTING DATA
Supervised Learning (Labeled)
Unsupervised Learning (Unlabeled)
Predictive AI (ML and DL)
Generative AI (Gen AI)
Multimodal Capabilities
COMPARING AI SUBSETS: SCALE & OUTPUT
CATEGORY
DATA SCALE
PRIMARY OUTPUT TYPE
EXAMPLE ENGINE
Machine Learning
1,000 - Millions
Prediction (Categories/Numbers)
Heart Disease Prediction
Deep Learning
Millions - Trillions
Prediction (Complex Patterns)
Neural Network Analysis
Generative AI
Billions (Parameters)
Generation (Text/Image/Video)
ChatGPT / Code
NotebookLM